still struggling with its 4G rollout, also wants airwaves for 5G so as to not miss out on the opportunity there.

India boasted a total mobile subscriber base of 1,143 million at the end of April 2022, according to data from the Telecom Regulatory Authority of India (TRAI)—many of whom could benefit immensely from the introduction of 5G services. Reliance Jio is the leader with 405 million subscribers, followed by Airtel (361 million) and Vodafone Idea (259 million). At the end of seven days of auctions on August 1, the Centre received bids worth over Rs 1.5 lakh crore from telecom operators. Reliance Jio topped the list with a bid of Rs 88,078 crore, followed by Airtel with Rs 43,084 crore, and Vodafone Idea with Rs 18,799 crore. Up for auction was about 72 gigahertz (GHz) of radio waves that can offer 5G services, worth around Rs 4.3 lakh crore. (Hertz is a frequency unit that indicates the speed at which information is processed over a second, and one GHz is equal to a billion hertz.) The government achieved 71 per cent of the targeted spectrum sales.

Significantly, the government has also allowed technology firms and corporates to set up captive 5G networks to enhance commercial offerings to their clients. The Adani Group is the first to go for it and has put in bids worth Rs 212 crore, indicating that it plans to start its telecom foray in a limited manner. Considering that it will take a few months for the 5G rollout, the services are expected to be implemented in phases. Urban areas with higher telecom tower density and a higher number of 5G handsets are likely to be the first to witness the implementation. "Our target is to start the rollout process sometime in early October and within one to one-and-a-half years, we should see a good presence of 5G in the country," Ashwini Vaishnaw, the Union minister of communications, electronics, IT and railways, told INDIA TODAY.

THE 5G REVOLUTION

Once this comes to pass, it will not just be in-home entertainment, 5G will bring about a revolution in our lives. Vaishnaw sees two major sectors being immediately impacted: health and education. In health, it could enable point-of-care diagnostics where patients from home can have video consultations with medical professionals and will not have to visit clinics unless necessary. It will also facilitate connected ambulance services (real-time data transfer to the hospital). In education, the minister says, given the vast services that 5G will offer, services will be redesigned around it rather than the other way around. He also points out that drone technology would get a major boost, especially in sectors like agriculture where it could be used for precision spraying of fertilisers and pesticides in addition to drip irrigation.

Amitabh Kant, the former CEO of NITI Aayog, in a recent article, listed other key areas where 5G would effect a transformational change for the common man. In the banking sector, he believes, the use of geospatial information systems (which create, manage, analyse and map all types of data) will allow us to reach the next level of simple, seamless and secure payments such as 'one-tap payment' (which eliminates layers of steps, making transactions hassle-free) and 'cashier-less store'. And that isn't all. In transportation and mobility, 5G can improve infrastructure for electric vehicles by identifying and communicating with charging stations and integrate initiatives across transit systems such as FASTag for toll and entry tax, potentially reducing traffic congestion. In manufacturing and industry, 5G would elevate automation efficiency to another level, reducing the carbon footprint. In governance and public safety, service delivery and citizen engagement could be improved with faster and safer digital verification.

To understand how 5G works, let's go back a bit in time. Even though baby steps toward mobile telephony were taken in the mid-20th century, it was not until the 1980s that the first commercial mobile telephony operators started gaining momentum. The year 1979 saw the introduction of 1G or the first generation of telecom services, enabling voice calls over mobile phones, which was followed by 2G in 1991 that offered improved sound quality, encrypted communications and SMS messaging. The launch of 3G in 2001 ushered in facilities such as global roaming, email and video streaming, offered for the first time on mobile phones, while 4G took them further into high-definition (HD) streaming, social media and interactive apps.



"EFFECTIVELY, **THERE WILL BE NO LATENCY OR DELAY BEFORE TRANSFER OF DATA**. THIS WILL MEAN IMMEDIATE CONNECTIVITY AND YOU CAN INTERACT IN AN IMMERSIVE WAY AS IF YOU ARE TALKING TO EACH OTHER"

–PRASHANT SINGHAL, *Technology, Media and Telecom Emerging Markets Leader, EY*